REMIXKIT · THE ARTIST CONSOLE

System architecture, and every trace a user leaves.

A record label registers an artist, builds that artist's identity once, attaches songs, measures each master, generates content designed to be imitated, approves it, and hands it off with the disclosure travelling inside the file.

This document maps that end to end: every surface a person touches, every document and object the touch writes, every claim the system makes about *how* it knows something, and the exact boundary of what taking it back can reach.

- **request / response** the ordinary synchronous path
- **durable bytes** everything that outlives the process
- **queued work** never inside an HTTP request
- **browser ↔ bucket** presigned; the bytes never enter compute
- **a vendor call** billed — and observably billed on failure too
- **provenance** the claim that travels inside the file
- **a refusal** on purpose, in the service layer, with a test

Generated by `docs/architecture/render.py` — labels carry the real routes, object keys and settings, so the picture cannot drift from the code.
Companion to `infra/architecture.pdf`, which is the AWS-shaped view of the same system.
Sheets are 18.75×12.5in — built to be read on screen at 100% and printed at A2.

The whole system on one sheet

One process, seven configuration axes, one bucket, no database. Read left to right for a request; read the purple lane for everything a request refuses to do itself.

a master is uploaded browser → bucket on a presigned PUT — the bytes never enter compute

WHO IS CALLING

Label operator
the only role in scope
- Browser + httpOnly **rk_session** cookie
- Registers, measures, previews, buys, approves

Public visitor
no session, ever
- Reads the landing page
- Checks a file at **/verify**

Script / integration
Authorization: Bearer
- Same services, same gates
- JSON in, JSON out

Worker principal
subject="worker"
- Minted inside the job, not by a login
- Carries the tenant off the

Every console handler asks for a `CurrentPrincipal`. That is the entire gate — no route contains an `if authenticated` branch.

GET + htmx

Bearer

FRONT DOORS

/ the landing page
public, always
the one route with no principal in its

/verify
public, on purpose
reads no tenant data — not under `/console`

/auth/login
six digits by email
sets the signed session cookie

/console/…
~20 screens · ~70 htmx fragment routes
- roster · artist · identity builder
- song · catalogue · settings
- **generate** — the priced dry run
- kit — polls one row every 3s

/api/v1/…
JSON · Bearer
- artists · identities · songs · sections
- lyrics · kits · verify · auth

/files/{key}
local storage only
never reached on B2 — and refuses anything under `tenants/`

ONE FASTAPI PROCESS · HEXAGONAL

ui/routes.py + api/v1.py
two routers, one set of services
a fragment route returns exactly the markup that changed

artists
roster · consent

identities
versions · frames

songs
masters · sections

recipes
the four formats

briefs
recipe → shots

kits
request() / run()

analysis
queue a measurement

transcription
queue a reading

recommendations
rank the hooks

delivery
embed on download

verify
read it back out

accounts
codes · sessions

catalogue
what is missing

ports/ — the eight protocols
services depend on these, never on a vendor
- Storage · DocumentRepository · Generator · JobQueue
- AudioAnalyzer · Transcriber · Mailer · AuthProvider

wired once, per process

deps.py — the composition root
the only file in the codebase that names a concrete adapter
"laptop with no credentials" and "Lambda + SQS + Batch + B2 + GMI" differ by seven environment variables

selects

ADAPTERS — CHOSEN IN DEPS.PY

RK_STORAGE_BACKEND
- dev storage_local
- prod **storage_b2**

RK_GENERATOR_BACKEND
- dev mock (ffmpeg pixels)
- prod **genblaze**

RK_QUEUE_BACKEND
- dev queue_inline
- prod **queue_sqs**

RK_MAIL_BACKEND
- dev mailer_console
- prod **mailer_zepto**

RK_AUTH_BACKEND
- dev anonymous
- prod **otp**

RK_ANALYSIS_BACKEND
- dev *no mock exists*
- prod **audio_numpy**

RK_TRANSCRIPTION_BACKEND
- dev *no mock exists*
- prod **lyrics_elevenlabs**

OUTSIDE THE BOUNDARY

Backblaze B2
every durable thing, one bucket
- `…/tenants/{t}/{collection}/{id}.yaml`
- `…/masters/{t}/{song}.{ext}`
- `…/tenants/{t}/identities/{id}/frames/…`
- `runs/{t}/{date}/{run}/…` + manifest

no database tier anywhere in this design

GMI Cloud
image today · video when unpinned
seedream-5.0-lite

OpenAI
video — pinned here today
sora-2

ElevenLabs
audio, and Scribe for lyrics
eleven_multilingual_v2 · scribe_v1

ZeptoMail
the six-digit codes

AWS SSM Parameter Store
read at boot, before settings are cached
a literal `PLACEHOLDER …` counts as unset

presigned GET back to the browser

THE ASYNCHRONOUS HALF — NOTHING SLOW HAPPENS IN A REQUEST

the fast half
request() — well under 200 ms
- validate · gate · write one document
- enqueue with **dedupe_key**

JobQueue
SQS in prod · inline on a laptop
- same code either side
- a queue that cannot execute refuses *before* a row is written

Dead-letter queue
after the redrive policy gives up

worker.py
AWS Batch · Fargate Spot · `minvCpus: 0`
- deliberately not a web process — Genblaze runs with `timeout=900`, which *is* Lambda's ceiling
- non-zero exit so a failure is recorded as one

run-kit
`--kit-id` · minutes · the expensive one

analyze-song
`--song-id` · a minute of CPU on a long master

transcribe-song
`--transcribe-song-id` · Scribe, direct

the document is rewritten
in place, same id
- status · assets · manifest_verified
- measurements · lyrics · cost

the console notices
polling one row
every 3s — no websocket, on purpose

idempotent on the document id — a redelivered message is a no-op, and a finished kit that arrives twice costs nothing

HOW TO READ THE WIRES

- **request / response** — the ordinary path
- **durable bytes** — everything that survives the process
- **queued work** — never inside an HTTP request
- **browser ↔ bucket** — presigned, never through compute
- **a vendor call** — billed, and possibly billed on failure
- **provenance** — the claim that travels in the file
- **a refusal** — on purpose, and tested

Onboarding, and everything the master is asked

Nine touchpoints. Read down a column for one action and everything it leaves behind; read across a lane to compare what the system keeps.

each station is reached from the one before it, but nothing forces the order — the console shows what is missing instead

	01 Sign in `POST /ui/auth/request-code → …/verify-code`	02 Register an artist `POST /ui/artists · POST /api/v1/artists`	03 Record likeness consent `POST /ui/artists/{id}/consent`	04 Build the identity `/console/artists/{id}/ident`	05 Attach a song `POST /ui/artists/{id}/songs`	06 Upload the master `…/master/upload-url → PUT → …/master`	07 Measure the master `POST …/songs/{id}/analysis`	08 Read the words off it `POST …/songs/{id}/transcription`	09 Set the hooks `POST …/sections · …/sections/{id}/primary`
what the operator touches the surface, and the control on it	**Types an email, reads a six-digit code, enters it.** · The console and the API share one token — the cookie is just where a browser keeps it.	**Name, slug, bio, links.** · An artist is a first-class entity, not a string on a song.	**Ticks *granted*, records who signed and when, links the release.** · Refusal #1 in the product lives here.	**Three columns at once: what the model is told, what it is shown, what was said before.** · chips for presentation · build · height, then face structure and wardrobe · reference frames, classed by six angles × lighting	**Title, slug, ISRC, Spotify link. A BPM only if it can say where it came from.** · Refusal #2 lives here.	**Drops an MP3, WAV, FLAC or M4A on the song page.** · The one console route that is not htmx — a presigned URL is JSON, and the browser does the PUT	**Presses *Measure*. The panel polls until it lands.** · Tempo, the drop, the riser, the plateau, and the sections — from the audio, not from a form	**Presses *Transcribe*, then fixes what the model misheard — one line, or the whole sheet in a textarea.** · Lines under 0.6 confidence are flagged **check by ear** rather than hidden	**Adds, moves and labels sections. Picks the primary hook. Reads the ranked recommendations and their basis.** · A song has as many hooks as it has — one of them is primary
what the server does service, and the gate it applies	**accounts — the allowlist *is* the user table** · `RK_ALLOWED_EMAILS` checked on request, on verify, **and on every session read** · A refused address is told so — membership leaks, deliberately · Hashed (HMAC, keyed by email) · single-use · 10 min · 5 attempts	**artists.create** · The tenant is created by the first request that mentions it · Four built-in formats are seeded on first read — there is no migration step to hang a fixture on	**artists.set_consent** · This is not validation — it is a product rule, in the service layer, with a test · PRODUCT.md gap #3 inverted: a label wants explicit, auditable rights so the face *can* be held	**identities.create_version** · Saving **mints a version** — within its own *line*, so a band's second member starts at v1 · `compile()` renders ≤ **220 chars** by salience and drops clauses **whole** · `plates()` picks which frames a run may send	**songs.create** · **No BPM without a method.** A field that silently accepts an unsourced 128 hides the real bottleneck — catalogue onboarding — instead of measuring it	**songs.presign_master / register_master** · The key is **derived**, never accepted: `{prefix}/masters/{t}/{so` `{ext}` · `register_master` re-checks the prefix — a caller naming any key could otherwise reach another tenant's object · and checks the bytes are actually there	**analysis → queued job `analyze-song`** · Two gates before the enqueue: a master whose bytes exist, and an analyser that can actually run · **There is no mock analyser.** A fabricated measurement is a float nothing downstream could tell from a real one · `worker.py --song-id …`	**transcription → `transcribe-song`** · ElevenLabs Scribe, direct — no generator step to hang it on · Words become lines on three stated thresholds: a 900 ms gap, 42 characters, sentence-final punctuation · Re-transcribing **replaces**; a song with hand-corrected lines refuses and says how many would be lost	**songs + recommendations** · `song.hook` is mirrored onto the primary section, so the two surfaces cannot disagree · Rules from RHYTHM-STUDY §9 report pass, fail, or **not checkable** — the third state matters · No weighted total — a score would read as a quality judgement
what is left behind the residue — documents, objects, fields	**Written** · `…/otp_challenges/{id}.ya` — code_hash, expires_at, attempts · `…/accounts/{id}.yaml` — last_login_at · **No session row.** The cookie is a signed, stateless token	**Written** · `…/tenants/{t}/artists/{a` · `approval: draft`, empty consent · `…/recipes/*.yaml` ×4, once per tenant	**Embedded on the artist** · `consent.granted / signed_by / signed_at` · `consent.document_key` — the signed release in the bucket · `consent.notes`	**Written** · `…/identities/{idn_…}.yam` — one document **per version** · `…/identities/{id}/frames/` — the key *is* the content digest · The full `sha256` is kept on the frame, because an unhashed plate makes the manifest move on every render	**Written** · `…/songs/{sng_…}.yaml` · `bpm` + `bpm_method`, or neither	**Written** · The master object — **uploaded browser → bucket**, never through compute · `song.master_key` + `master_content_type`, recorded *after* the bytes land	**Written onto the song** · `analysis` — bpm · beat_ms · downbeat_ms · drop_ms · riser_beats · plateau_beats · bar_phase_confidence · `analysis.bars[]` — per-bar spectra, and a printable sheet · `sections[]` with `source: measured`	**Written onto the song** · `lyrics` — language, confidence, status · `lyrics.lines[]` — text, window, per-line confidence · **There is no mock transcriber** — a fabricated lyric is words attributed to an artist	**Written onto the song** · `sections[]` — role · window · bar_start/end · evidence · `primary_section_id` · Nothing is written by asking for a recommendation — it is computed on read
what is recorded about *how* provenance: the claim, and its basis	**Who, and when.** · `last_login_at` on the account · Removing an address ends the session at the next request, not at its expiry	**`tenant_id` is on the document, in the object key, and on the job payload.** · One tenant axis, everywhere — the thing BUILD-SPEC calls near-impossible to retrofit	**The audit record is the two fields.** · `signed_by` + `signed_at` are what a rights question is answered from	**Version, angle, lighting, digest.** · `approval` per version · The compiled string, its length against the budget, and a count of what was refused are all on screen	**The method line is the price of the number.** · FORMAT-SPEC requires the provenance of a measurement; this is that requirement in code	**An abandoned upload leaves no claim.** · The key is deliberately not recorded at presign time — a song claiming a master it does not have reads as "ready to analyse" on every screen that asks	**Which adapter, how, and on which bytes.** · `analyzer`, `method` verbatim from the adapter, `source_key`, and warnings	**The thresholds are in the stored method line.** · Each line carries the geometric mean of its words' probabilities · Editing a line makes it `manual`, keeps the old method as history, and **drops the confidence**	**Every window records where it came from.** · Moving a `measured` window makes it `manual` and keeps the old method as history · A dragged window still reading "measured" is that rule defeated by the edit
what undoing it costs reversibility, and what it cannot reach	**No revocation before expiry.** · TTL is a week. The panic button is rotating `RK_SESSION_SECRET`, which signs everybody out at once	**Delete refuses while anything points at them.** · It names the counts. `cascade` is the caller saying yes — and cascading runs *through* the owning services so each cleans its own objects	**Withdrawal is as easy as granting —** · and takes effect on the **next** kit, never retroactively. Kits already rendered keep their assets	**Full CRUD, with two rules that are easy to get wrong.** · Removing a frame deletes its object **only if** no sibling shares the key · Deleting a version deletes only unreferenced objects · Restore **copies forward**: it	**Delete removes the master object too,** · and refuses while a kit still references the song	**Re-uploading reuses the same key.** · Deleting the song deletes the object	**Re-running replaces the measured sections.** · Idempotent in the way that matters — a redelivered message costs a minute of CPU and changes nothing	**Editing is free; re-reading is not.** · `force` is the caller saying yes anyway, and the console asks first	**Freely editable —** · but a kit already bought stores *its own copy* of the windows, so editing cannot move what a finished kit claims

Two of these stations refuse by design — consent (03) and an unsourced BPM (05) — and both refusals live in the service layer, not in a form validator.

Buying a run, and what leaves the building

Nine more touchpoints. Two of them write nothing at all — the dry run and the public verifier — and one of them is where every dollar in the system is spent.

the operator is only present for five of these nine — the rest is a worker, a manifest, and a file that carries its own record

	See the run before buying it 10 `GET …/songs/{id}/generate`	Press the queue button 11 `POST /ui/kits · POST /api/v1/kits → 202`	The queue, and the worker 12 `SQS → AWS Batch (Fargate Spot)`	The run 13 `Genblaze Pipeline → providers`	The manifest, and the gate 14 `manifest.verify()`	Review and approve 15 `POST /ui/kits/{id}/approval`	Download 16 `GET …/assets/{id}/download`	Verify — the public door 17 `POST /api/v1/verify · POST /ui/verify`	Delete, and withdraw 18 `DELETE /ui/… or /api/v1/…`
what the operator touches the surface, and the control on it	**Picks the format, the faces, which hooks, how many, and edits any shot's prompt, model, length or lock.** · Every control re-prices the page on change — htmx, still a GET to the same handler · The estimate sits outside the scroll — it must not	**Commits the plan that is on the screen.** · Everything that alters the plan rides the button — a control that changes what is bought without changing what was priced is the divergence this screen exists to prevent	**Nothing. The operator is on another page.** · The kit page polls one row every three seconds	**Nothing — but this is where the money goes.** · A submitted step is a charged step. The only safe place to refuse was station 11	**Nothing.** · This is the step that decides whether the kit is allowed to exist	**Watches the row fill in, opens the assets, and decides.** · A label does not auto-publish AI content about its own artists	**Takes the file.** · This is the moment the content leaves the system	**Anyone drops the file back in. No account.** · A judge with no session can check a claim end to end	**Removes an artist, a song, an identity version, a frame, a kit — or withdraws consent.** · The console asks by *name and count* rather than behind a generic "are you sure"
what the server does service, and the gate it applies	**kits.plan → `_resolve()`** · **One resolution path.** `plan()` displays the payloads and `generate()` submits them — one function, two callers · Resolves the model that would answer, the composed prompt as the provider receives it, the negatives, the snapped duration and the price · Sends nothing	**kits.request — four gates, in order** · **1** No recorded likeness consent → refused · **2** A queue that cannot execute → refused before a row is written · **3** Estimate over the kit budget or `RK_MAX_RUN_CENTS` → refused · **4** A brief that produces no shots → refused	**worker.py — one job, then exit** · Deliberately not a web process: Genblaze runs with `timeout=900`, which is exactly Lambda's ceiling with no headroom · `minvCpus: 0` — nothing runs, and nothing is billed, at rest · Exits non-zero, so a failure is recorded as one	**generator_genblaze** · The compiled identity is **prepended to every prompt**; negatives ride as `negative_prompt` · **Two-stage identity lock:** a plate-conditioned still, then the clip generated from it via `input_from` · A still already in the index is **reused free** — nothing submitted, nothing billed	**kits.run — the readiness rule** · `ready` requires assets **and** a verified manifest · An unverified manifest means the provenance claim is unsupported — so the kit **fails** rather than shipping a claim that cannot be backed	**kits.set_approval** · The human gate — a field, moved deliberately, by a person · The same gate exists on artists and on identity versions	**delivery — embed on delivery, not before** · The run's manifest is written **into the media itself** · `X-RemixKit-Provenance` says which happened · A media type with no embedding handler is delivered *unmodified* and the caller is told — silently handing back an undisclosed file is the one failure this service exists to	**verify** · Reads the manifest *out of the bytes* · Returns `verified: true`, `source: "embedded"`, and the provider, model and prompt of every step · **Nothing is looked up.** It reads no tenant data at all	**Refuse, then cascade on request.** · Deleting an artist with dependents is refused and the counts are named · `cascade` runs **through the owning services**, so each dependent cleans up its own bucket objects · Silent orphaning is the worst of the three options
what is left behind the residue — documents, objects, fields	**Nothing.** · The only station in the product that writes no document and no object · That is what makes it safe to open — and why three defects stayed invisible while no screen showed the payload	**Written** · `…/kits/{kit_…}.yaml`, `status: queued` · and the **brief** — recipe_slug · hook windows *as bought* · shot_overrides by index · identities by id, name *and* version · estimate_cents · which generator · One SQS message, `dedupe_key = kit.id`	**Rewritten** · `kit.status: queued → running`, then `ready` or `failed` · A message that outlives its retries lands in the DLQ · CloudWatch logs, 14-day retention	**Written** · Assets under `runs/{t}/{date}/{run_id}/` — HIERARCHICAL, tenant as a path component · `LockedStill` appended to the *identity line*, not to the kit — its whole value is outliving the kit that paid for it · Every step's provider, model, prompt, params and cost	**Written** · The manifest object, beside the assets · `kit.run_id · manifest_key · manifest_verified · assets[] · total_cost_cents` · The stored object stays **byte-identical** to what the provider returned	**Rewritten** · `approval: draft → approved` · `updated_at`	**Nothing new in the bucket.** · The stored object is untouched — that is what makes its content hash mean anything · The residue leaves with the operator: a file carrying its own record of which model made it, from which prompt, for which release	**Nothing.** · No document, no object, no session · Which is exactly why the route is not under `/console` — that prefix promises a session check this handler does not perform	**What survives is what something else references.** · A frame's object survives if a sibling frame shares its key · A version's objects survive if any surviving version references them · Withdrawn consent leaves rendered assets alone
what is recorded about *how* provenance: the claim, and its basis	**The wire payload, on screen, before the money.** · A truncated mood fragment once reached the provider naming no song, no artist and no constraint against audio — and nothing had "failed"	**The brief freezes what was quoted.** · Windows are copied, not referenced — an edited section cannot change what a finished kit claims it was cut to · Faces by id **and** name, so "which member" needs no	**Idempotent on the document id.** · A finished kit that is redelivered is a logged no-op and costs nothing	**Identity versions are inside the still digest.** · Saving a new version must *miss* the index rather than reuse a frame of how that person used to look · The scene is in the digest too, so a night-drive still is not offered for a golden-	**Unknown and free are different claims.** · `cost_cents: int | None` — an unpriced model estimates at `UNKNOWN_CALL_USD`, never at zero · `cost_is_complete` shows a floor, not an invoice	**Approval is a state on the record,** · not an event log — this scope keeps no audit trail beyond the fields on the documents	**Disclosure now travels inside the file.** · Nothing has to be looked up, and nothing can be quietly stripped without the check failing	**This is the loop closing.** · Generate → verify → deliver embedded → read back. Tested as a loop, not as three assertions	**Neither rule errors when it is wrong.** · Get them backwards and the rows remain while the images 404 — the shape of data loss that is hardest to notice
what undoing it costs reversibility, and what it cannot reach	**Nothing to undo.** · Clearing an edited field restores the generated default rather than sending an empty prompt	**Deleting the kit removes the bucket objects it owns,** · not just the document. Orphaned assets are storage nobody can reach	**A failed kit keeps its error on the document** · and can be deleted; there is no partial resume	**Assets are content-addressed.** · A re-run mints a new `run_id`; it does not overwrite the last one	**A failed manifest is not retried silently.** · The error is on the document and on the screen	**Freely moved back.** · Approval gates publication, not generation — the money was already spent at 13	**Unreachable.** · A downloaded file is gone. Deleting the kit later removes the stored copy, not the delivered one	**n/a** · Strip the manifest and the check fails. That is the point	**Nothing here is undoable.** · Identity *versions* are the exception — restoring an old one copies it forward as a new one

Station 10 is the whole cost-control argument: there is one resolution path, so the payload displayed before the button is the payload submitted after it.

Everything a user leaves behind, and what it costs to take back

The product keeps no event log. Every claim it makes is a field on a document beside the sentence explaining where the field came from — so this page is the audit trail.

ONE BUCKET · THE WHOLE KEY SPACE

```
remixkit/                                          ← RK_KEY_PREFIX
├── tenants/{tenant_id}/                           ← the partition key, in the path
│   ├── artists/{art_…}.yaml                       Artist · consent · approval
│   ├── identities/{idn_…}.yaml                    one document per version
│   ├── identities/{idn_…}/frames/
│   │       {sha256[:16]}.png                      the key is the content digest
│   ├── songs/{sng_…}.yaml                         sections · analysis · lyrics
│   ├── kits/{kit_…}.yaml                          brief · assets · ledger
│   ├── recipes/{rcp_…}.yaml                       four, seeded per tenant on first read
│   ├── accounts/{id}.yaml                         last_login_at
│   └── otp_challenges/{id}.yaml                   code_hash · expires_at · attempts
│
├── masters/{tenant_id}/
│       {sng_…}.{wav|mp3|flac|m4a}                 browser → bucket, presigned PUT
│
└── runs/{tenant_id}/{date}/{run_id}/
        …assets…                                   byte-identical to the provider's output
        manifest.json                              the claim, and its canonical hash
```

Three prefixes, one tenant axis. Documents, masters and run output share it, so a tenant is a path component rather than a column — the thing BUILD-SPEC §2b calls near-impossible to retrofit. `/files/{key}` exists only on local storage, and refuses anything under `tenants/` outright: the YAML is the database, not a file to be served.

REVERSAL · WHAT EACH UNDO CAN AND CANNOT REACH

Cascading goes through the owning services, never through the repository. That is the only reason each dependent cleans up its own bucket objects. Deleting the rows directly is the cheap implementation and the one that leaves the label paying storage for objects no screen can reach.

WHAT A USER CAN UNDO	WHAT IT REACHES	WHAT IT CANNOT
Withdraw likeness consent	the **next** kit — refused at request time	anything already rendered, delivered or downloaded
Delete an identity version	objects *no surviving version* references	a kit that recorded that version in its brief
Remove a reference frame	its object, if no sibling frame shares the digest	a still already in the locked-still index
Edit a section window	the song, and every future kit	a finished kit — it stored its own copy of the windows
Delete a kit	the document, its assets and its manifest	a file already downloaded, which carries its own manifest
Rotate `RK_SESSION_SECRET`	every session, at once	nothing else — sessions are signed, not stored

EVERY DURABLE RECORD, AND THE RESIDUE ON IT

DOCUMENT	WRITTEN WHEN	THE RESIDUE THAT MATTERS	PROVENANCE CARRIED
Artist	registration	slug · name · links · **consent** · approval	`signed_by`, `signed_at`, `document_key` of the release
Identity *(one per version)*	every save	line name · version · silhouette enums · wardrobe · negatives · **reference_frames[]** · **locked_stills[]**	version within its line · per-frame `angle`, `lighting`, `sha256`
Song	attach, then measured / heard	bpm · drop_ms · **sections[]** · **analysis** · **lyrics** · master_key	`bpm_method` · per-section `source` + `method` · per-line `confidence`
SongAnalysis	the worker finishes	bpm · beat_ms · downbeat_ms · drop_ms · riser_beats · plateau_beats · bar_phase_confidence · bars[] · sheet	`analyzer` · `method` verbatim · `source_key` · warnings
SongLyrics	the worker finishes	language · lines[] with windows · status	`transcriber` · the three grouping thresholds, in the method line · per-line confidence, dropped on edit
Recipe	seeded per tenant, then copied	prompt_template · negatives · aspect · length policy · face policy · variants	`builtin` — the four that refuse to be deleted
Kit	the queue button	status · **brief** · assets[] · manifest_key · **manifest_verified** · total_cost_cents · approval	run_id · hook windows *as bought* · identities by id + name + version · estimate at purchase
Asset *(on the kit)*	each provider call returns	modality · key · **cost_cents: int | None** · params	`provider` · `model` · `prompt` · `reference_keys` · `derived_from`
Account / **OtpChallenge**	sign-in	email · last_login_at · code_hash · attempts	the allowlist is the user table — there is no registration

WHAT THE RESIDUE IS FOR

CLAIM	WHAT IT RESTS ON	AND WHAT WOULD DEFEAT IT
“this window was measured”	`section.source = measured` plus the analyser's own method line	an edit form that leaves the label alone — so moving a measured window makes it `manual` and keeps the old method as history
“this lyric is what the model heard”	per-line confidence, and lines under 0.6 shown as **check by ear**	hiding the low-confidence lines, which makes a transcript look finished when the number says it is not
“this is the same person as last time”	the identity *version* in the still digest	a cache keyed on the face alone, which would quietly reuse a frame of how they used to look
“this kit was cut to that hook”	the windows copied into the brief at purchase	storing section ids, which lets a later edit change what a finished kit claims
“this clip cost 42¢”	one dated price book registered into Genblaze's own registry, read by both the estimate and the ledger	a flat per-modality table — which quoted $1.26 for a run that charged several dollars
“this file was made by that model”	the run manifest embedded in the media at delivery, verified before the kit was allowed to be ready	embedding at generation time, which would break the content hash the manifest is built on
“nothing was billed for that still”	an index hit: the plan row reads *reusing a stored still, free* and no step is submitted	a cache that reports a hit and submits anyway

The pattern: every measured value is stored next to the sentence that says how it was measured, and every human edit moves the value's provenance to `manual` while keeping the old method as history. A number without its method is refused at the door — that is refusal #2 — and a number whose method survived an edit that invalidated it would be the same defect, arriving later.

Nothing here is derived from a database schema: there is no database. These are YAML documents in one bucket, and the key layout is the index.

One resolution function, and the two-stage lock around a face

Everything between a brief and a billed provider call. The load-bearing property is that the dry run and the real run are the same function — so the payload on screen is the payload on the wire.

INPUTS

Recipe — a format is a row
four ship; copies are editable, built-ins refuse deletion

- prompt template · negatives · aspect · length policy
- face policy · which reference class the framing wants · variant spread
- **default is `performance`**, not the backdrop — the backdrop deliberately has nobody in it

Song
measured once, reused forever

- the chosen sections' windows — each loop is the length of *its own* window
- videos are dealt **round-robin** across hooks, not multiplied by them

Identity lines
a band is one artist with four faces

- `compile()` → ≤ 220 chars, ordered by salience, clauses dropped **whole**
- `plates()` → duplicates dropped by hash, one frame per lighting setup
- each face composes under its own name — two unlabelled silhouettes read as one contradictory person

What the operator typed
stored as `brief["shot_overrides"]`

- per shot: prompt · model · seconds · whether the face is locked at all
- kept as a list by index — JSON has no integer keys, and `{0:…}` comes back as `{"0":…}` matching nothing

The ceilings
checked before anything is submitted

- `RK_MAX_RUN_CENTS` · the kit's own budget · `RK_MAX_SHOTS_PER_KIT`
- Genblaze documents no spend guardrail, and a failed step is observably billed — so a submitted step is a charged step

RESOLUTION — ONE FUNCTION, TWO CALLERS

briefs.plan_from_recipe() → ShotSpec[]
then `apply_overrides()`

- one planner for the preview and for the run — not two implementations kept in step by hand

GenblazeGenerator._resolve()
the single resolution path

- **provider** — pins first, then keyed availability: GMI → OpenAI → Google (video/image), ElevenLabs → OpenAI (audio). A modality nobody serves is skipped with a warning
- **model** — follows the provider, from `DEFAULT_MODELS`. Defaulting to GMI slugs was a guaranteed 404 the moment anything else answered
- **duration** — snapped, so the quote prices the clip that will actually render
- **price** — one dated book, registered into Genblaze's own ModelRegistry

Will the reference actually be read?
two checks, at two levels

- `accepts_chain_input` — the *provider* takes an image at all. Genblaze raises before any step runs, so an ungated plate fails the whole kit
- `honours_reference()` — the *model* reads it. **An unknown model answers no**: wrongly assuming image-capable renders a plausible stranger and bills for it
- A shot that wants the face **swaps the model** to `RK_IMAGE_EDIT_MODEL` — it does not drop the reference. There is no default, on purpose

adapters/model_families.py
payload shapes the connector does not ship

- every entry is a rejection that named its own missing parameter — `bria-fibo-edit → 400: images (Required parameter is missing)`

plan() displays them · generate() submits them
one function, two callers — which is the whole cost-control argument

submit

EXECUTION

Genblaze Pipeline
one Run · steps fan out to `RK_MAX_CONCURRENCY` · timeout 900 s

- the compiled identity is prepended to every prompt; negatives ride as `negative_prompt`

Stage one — the still
conditioned on the artist's reference plates

- roughly a fifteenth of the clip's cost
- the expensive, uncertain part happens once, cheaply, where a person can look at it
- a video shot takes exactly **one** plate — a second lands in `last_frame` and reads as "end the clip here"

store

The still index
`still_digest(identities, prompt)` on the identity line

- a hit skips the image call entirely — nothing submitted, nothing billed
- it lives on the *line*, not the kit, because its whole value is outliving the kit that paid for it
- identity **versions** are in the digest, so a new face misses rather than reusing an old one

or reuse, free

Stage two — the clip
generated from that still via `input_from`

- a locked clip carries **no** plates of its own — positional slots would put two different images in first_frame and last_frame and ask the model to interpolate
- an index hit conditions through `external_inputs` instead

It degrades, it does not fail
and the plan row says which applies

- no image provider · no reference frames · a provider that refuses images → a plain plate-conditioned clip, with the reason on the row
- local storage with a live backend → refused, because the provider fetches the image from its own network and `/files/…` means nothing off-host
- a duo today sends one plate and *says so* — silent conditioning on one member is the failure that must not be invisible

OUTPUT, AND THE GATE

ObjectStorageSink · HIERARCHICAL
`runs/{tenant}/{date}/{run_id}/…`

- the tenant is a path component, so the bucket, the documents and the manifests share one tenant axis

The manifest
every step: provider · model · prompt · params · cost

- `manifest.verify()` is the gate, not a decoration
- assets **and** a verified manifest, or the kit is `failed`

The ledger
`cost_cents: int | None`

- unknown pricing estimates at `UNKNOWN_CALL_USD`, never at zero — coercing None to 0 is how a multi-dollar run reported $0.00
- `cost_is_complete` shows a total as a floor rather than as the invoice
- both stages are priced: one shot bought, two calls charged

The kit document, rewritten
in place, same id

- run_id · assets[] · manifest_key · manifest_verified · total_cost_cents · error
- `queued → running → ready | failed`
- a redelivered message finds it `ready` and does nothing

And back onto the identity
`LockedStill` appended to the line

- digest · key · prompt · identity_ids · approved
- this is the one write in the whole run that lands outside the kit — and it is why the *second* kit for an artist is cheap

the index the next kit hits

WHAT ACTUALLY ANSWERS, AND WHAT IT COSTS

MODALITY	DEPLOYED TODAY	ORDER WHEN UNPINNED	WHY
Video	**Sora** *sora-2* — **pinned**	GMI Cloud → OpenAI → Google	GMI's `seedance-2` failed twice after ~174 s and billed for output it never produced. The fix had to be a deploy-time setting: a provider misbehaving in production is an operational event, not a code change
Image	**GMI Cloud** *seedream-5.0-lite*	GMI Cloud → OpenAI → Google	cheap, and it works. But it is *text-to-image* — so any shot that wants the face swaps to `RK_IMAGE_EDIT_MODEL` rather than sending a reference nothing consumes
Audio	**ElevenLabs** *eleven_multilingual_v2*	ElevenLabs → OpenAI	the same key Scribe reads for transcription — a keyed deployment gains lyrics without gaining a vendor

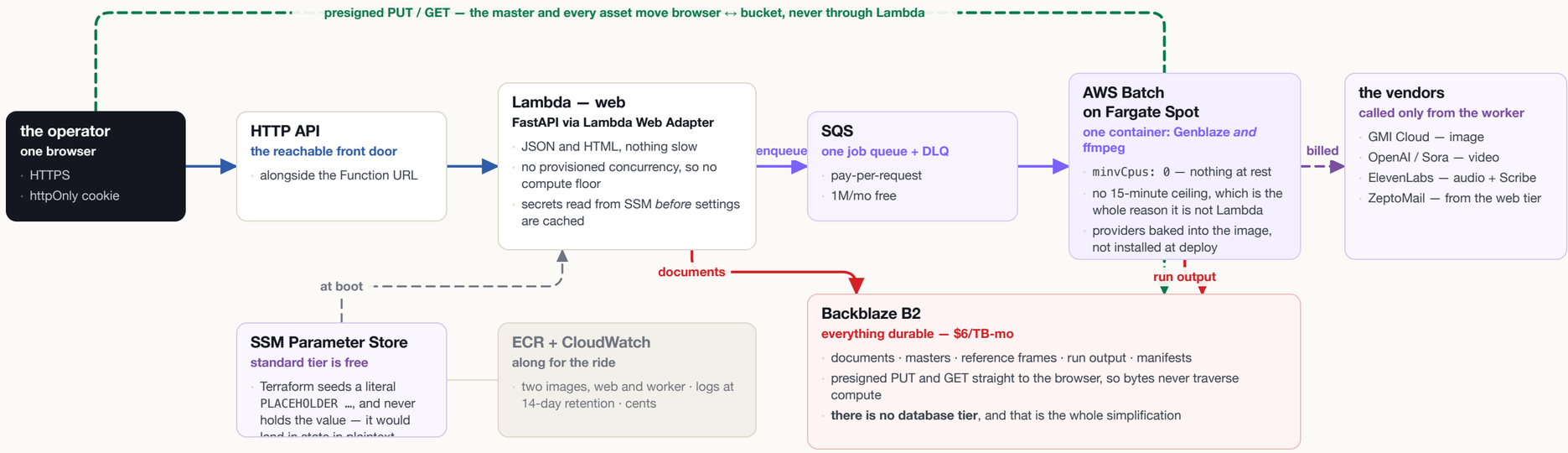
A REFUSAL YOU WILL MEET	WHAT IT MEANS
`seededit-3-0-i2i-250628` estimates at $2.00/call	it is unpriced, so it takes the unknown-model rate. Three locked clips quote at 783¢ and are refused by the 500¢ ceiling — reconcile the real rate against an invoice and register it. Until then, **the refusal is the system working**
"no likeness held — set `RK_VIDEO_EDIT_MODEL`"	both models between a photograph and a clip can silently discard the reference, and the connector declares `first_frame` for every `seedance-*` slug — true of 1.x, not of 2.0. Nothing is guessed: a hardcoded slug 404'd once already
"2 faces — 1 reference sent"	Genblaze routes any number of images, but GMI ships no per-model payload schema and extras are dropped. One reference is the only count this code can honestly claim, so it is the default
a silent soundtrack that is not silent	`negative_prompt` is a conditioning signal; no provider here documents it as binding on audio. The deterministic fix is dropping the track on delivery, and it is specified, not built

Prices, provider order and payload shapes are facts about vendors, not about this code: each one here arrived as a rejection or an invoice, and is registered rather than inferred.

Five services, one durable store, and no database tier

The same code runs on a laptop with zero credentials and on Lambda + SQS + Batch against B2 and three vendors. Seven environment variables are the entire difference.

PRODUCTION — AWS RUNS THE LOGIC, B2 HOLDS THE BYTES



Realistic idle cost: under $1/month — B2 storage and a container image. There is no compute floor anywhere, and the reason is that the relational schema this product would otherwise need exists almost entirely to serve *fans*: attribution joins, leaderboards, click ledgers, reward rows. Those are deferred, and the pressure for a database tier goes with them. What the label actually stores is documents, and a bucket is excellent at documents.

THE LAPTOP, WITH NO CREDENTIALS

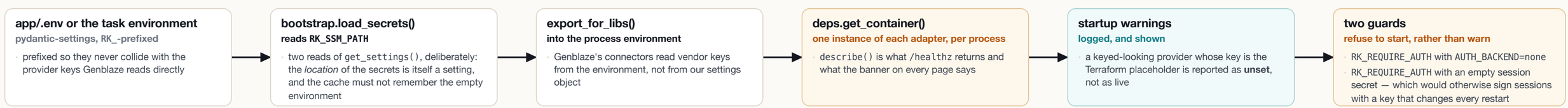
AXIS	DEV	PRODUCTION
`RK_STORAGE_BACKEND`	local	b2
`RK_GENERATOR_BACKEND`	mock	genblaze
`RK_QUEUE_BACKEND`	inline	sqs
`RK_MAIL_BACKEND`	console	zeptomail
`RK_AUTH_BACKEND`	none	otp
`RK_ANALYSIS_BACKEND`	*refuses*	auto (numpy)
`RK_TRANSCRIPTION_BACKEND`	*refuses*	auto (Scribe)

Not one line of application code differs between the two columns — services take ports, and only `deps.py` names an adapter.

Two axes have no mock, deliberately. A mocked *measurement* is a float nothing downstream could tell from a real one; a mocked *lyric* is words attributed to an artist. Both refuse and name what is missing — on the song page and in `/healthz`.

What "mock" does not mean. Real: the Pipeline, the sink, the key layout, content hashing, the manifest, `manifest.verify()`, the cost ledger and embed-on-delivery. Synthetic: the pixels, which ffmpeg draws, and the unit costs, which come from a table. So the manifest verified on a laptop is produced by the same code that produces the one in B2 — which is the only reason a zero-credential demo is worth having.

BOOT, CONFIGURATION, AND THE TWO GUARDS THAT REFUSE TO START



TURNING AUTHENTICATION ON	WHY IT IS SHAPED THIS WAY
`RK_AUTH_BACKEND=otp`	the default `none` is a complete implementation that admits everyone, so a fresh checkout has no login to get past
`RK_ALLOWED_EMAILS=…`	**this is the user table.** No registration. Checked on request, on verify, and on every session read
`RK_SESSION_SECRET=…`	sessions are a signed token, not a row — so there is no revocation before expiry, and rotating this is the panic button
`RK_MAIL_BACKEND=zeptomail`	with no mailer the code goes to the log, and back to the caller *only* in `RK_ENV=dev` — either condition alone would be a bypass

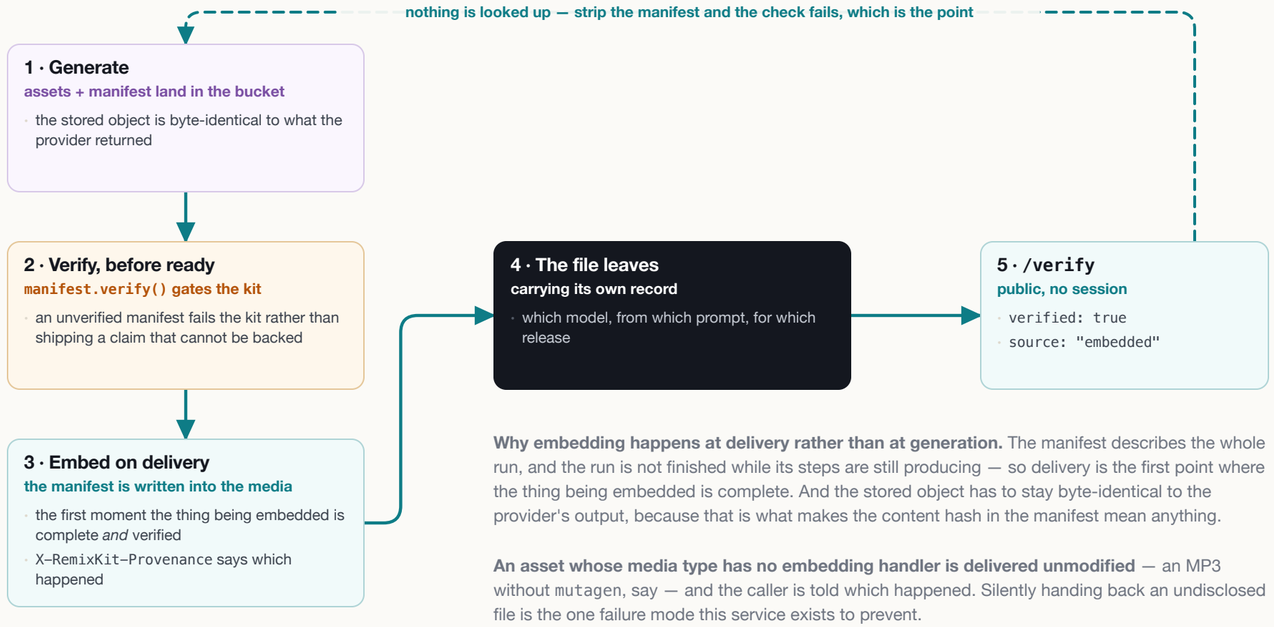
WHAT ADDING AUTH COST	BECAUSE
one provider, one service, one page, one exception handler	every request already resolved a `Principal`, and every document, object key and job payload already carried its `tenant_id`
no service changed	services take ports; routes take services; only `deps.py` names an adapter
no template and no storage key changed shape	the tenant axis was already in the path — "componentized, and we add auth later" became a configuration change rather than a refactor
no route grew an `if authenticated` branch	one `AuthError` handler decides what a failure looks like: a browser is redirected with `?next=` preserved, an htmx request gets `HX-Redirect`, a script gets 401 JSON

Terraform holds the shape but never the secrets — each parameter is seeded with a literal placeholder, and the provider table treats that placeholder as unset.

Fifteen refusals, and one loop that closes

The claim is that disclosure travels inside the file. Everything on this page exists so that the claim is either true or loudly absent — never quietly assumed.

THE PROVENANCE LOOP — TESTED AS A LOOP, NOT AS THREE ASSERTIONS



Why embedding happens at delivery rather than at generation. The manifest describes the whole run, and the run is not finished while its steps are still producing — so delivery is the first point where the thing being embedded is complete. And the stored object has to stay byte-identical to the provider's output, because that is what makes the content hash in the manifest mean anything.

An asset whose media type has no embedding handler is delivered unmodified — an MP3 without `mutagen`, say — and the caller is told which happened. Silently handing back an undisclosed file is the one failure mode this service exists to prevent.

WHAT IS DELIBERATELY NOT HERE

ABSENT	WHY
attribution links · `/r/{code}` · the disclosure gate · leaderboards · rewards · composite sessions · K-factor	all of it serves the *fan* role, which this scope defers. They are also what would force a relational database — their absence is why there is no database tier
a websocket	the kit page polls one row every three seconds; a viral release exercises storage reads, not sockets
a CDN	there is no read path hot enough to justify one at this scope
a “which archetype goes viral” recommender	research calls that folklore. The song page ranks hooks against stated rules and reports **not checkable** when it cannot — there is no weighted total anywhere, because a single score would read as a quality judgement the material cannot support
an event log or audit trail	provenance is a field on the document beside the value it describes. That is a deliberate trade: it answers “how was this measured” perfectly and “who changed it last Tuesday” not at all

EVERY REFUSAL IN THE PRODUCT, AND WHAT IT IS PROTECTING

#	THE REFUSAL	PROTECTING
01	No recorded likeness consent → **no kit**	a kit holds the artist's face invariant across every shot, and that needs auditable rights. Withdrawal takes effect on the next kit, not retroactively
02	A BPM with no method → **refused**	a field that silently accepts an unsourced 128 hides catalogue onboarding, which is the actual bottleneck
03	Estimate over `RK_MAX_RUN_CENTS` or the kit budget → **refused before submission**	Genblaze documents no spend guardrail and a failed step is observably billed, so the only safe assumption is that a submitted step is a charged step
04	A queue that cannot execute → **refused before a row is written**	otherwise a kit sits at `queued` forever with nothing coming for it
05	Assets without a verified manifest → **the kit fails**	the disclosure argument *is* the product; shipping an unbacked claim is worse than shipping nothing
06	No numpy → **the analyser refuses and names what is missing**	a mocked measurement is a float nothing downstream could tell from a real one
07	No credentialed transcriber → **refused**	a fabricated lyric is words attributed to an artist, offered to the generate form and burned into a clip
08	Re-transcribing over hand-corrected lines → **refused unless forced**	a lyric is one continuous reading; interleaving old corrections with new lines produces a transcript that is neither
09	Deleting an artist with dependents → **refused, with the counts named**	silent orphaning is the worst of the three options — a roster that looks clean while the catalogue counts songs nobody can open
10	A live generator on local storage → **the plate is refused**	the provider fetches the image from its own network, and `/files/…` means nothing off-host. Sending a dead link would look like it worked
11	An unknown image model → **treated as not reading references**	the safe direction: a model wrongly treated as image-capable renders a plausible stranger and bills full price for it
12	`RK_IMAGE_EDIT_MODEL` unset → **the plan says no likeness will be held**	there is no default because a hardcoded slug 404'd once already, and GMI ships no catalog endpoint to check one against
13	A master key that is not this song's → **refused**	it arrives from a browser; a caller who could name any key could point a song at another tenant's object
14	An email outside the allowlist → **told so**	this leaks membership, deliberately: with a handful of known colleagues the real failure is somebody mistyping their own address
15	`RK_REQUIRE_AUTH` without a backend or a secret → **the process refuses to start**	a warning here would mean a deployment that looks authenticated and is not

Each refusal is in the service layer with a test, not in a form validator — these are product rules, and a rule enforced only by a screen is a rule the API does not have.